



UNIVERSITI TUN HUSSEIN ONN MALAYSIA

**TEST
SEMESTER I
SESSION 2024/2025**

- COURSE NAME : DATA STRUCTURE
- COURSE CODE : BIC 10404
- PROGRAMME CODE : BIP/BIM/BIS/BIW
- EXAMINATION DATE : NOVEMBER 2024
- DURATION : 2 HOURS
- INSTRUCTIONS :
1. ANSWER ALL QUESTIONS
 2. THIS TEST IS CONDUCTED VIA
 - ☐ Open book
 - ☐ Closed book
 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

THIS QUESTION PAPER CONSISTS OF **FOUR (4)** PAGES

Q1 A human resource manager is assigned to report performance of a production team with twelve employees. The manager has to record each employee's ID, job position, basic salary, and total working hours per week.

(a) Define the structure for the employee record.

(5 marks)

Answer:

(b) Declare an array that is meant to store data for the defined structure in **Q1(a)**.

(1 mark)

Answer:

(c) Write a code segment that will allow the trainer to enter data for each employee into the array declared in **Q1(b)**.

(6 marks)

Answer:

(d) Write a function that will calculate and return the average total working hours per week for all employees in the production team.

(10 marks)

Answer:

Q2 Questions **Q2(a)**-**Q2(e)** are based on the following program:

```
#include <stdio.h>
#include <stdlib.h>

struct Number{
    int data;
    struct Number *next;};

struct Number *front=NULL, *p1;

int main()
{
    int n=1;
    while (n<=10)
    {
        p1=malloc(sizeof(struct Number));

        if (n%2!=0)
            p1->data=n*5;
        else
            p1->data=n*2;

        p1->next=NULL;

        if (front==NULL)
            front=p1;
        else
        {
            p1->next=front;
            front=p1;
        }

        n++;
    }
    return 0;}
```

- (a) Write a function to display the content of the linked list in a single line. (10 marks)

Answer:

- (b) Write a statement to invoke the function in **Q2(a)**.

(1 mark)

Answer:

- (c) Show the output when the statement in **Q2(b)** is executed.

(11 marks)

Answer:

- (d) Write a function to find and return the largest data from the linked list.

(16 marks)

Answer:**- END OF QUESTIONS -**